UNIT : 3

Linear Programming :

Model :-

- → objective function.. (max z , min z)
- → constraint ($\geq$, $\leq$) conditions.

Ex:-

$$\max\ z = x + 2y$$

$$2x + 3y \geq 5$$
$$x + y \leq 7$$
$$\boxed{x, y \geq 0}$$

$$\min\ z = 2x + 3y$$

$$4x + 3y \leq 8$$
$$2x + y \geq 4$$
$$\boxed{x, y \geq 0}$$

Graphical Method :

Ex:- $\max\ z = 3x_1 + 4x_2$

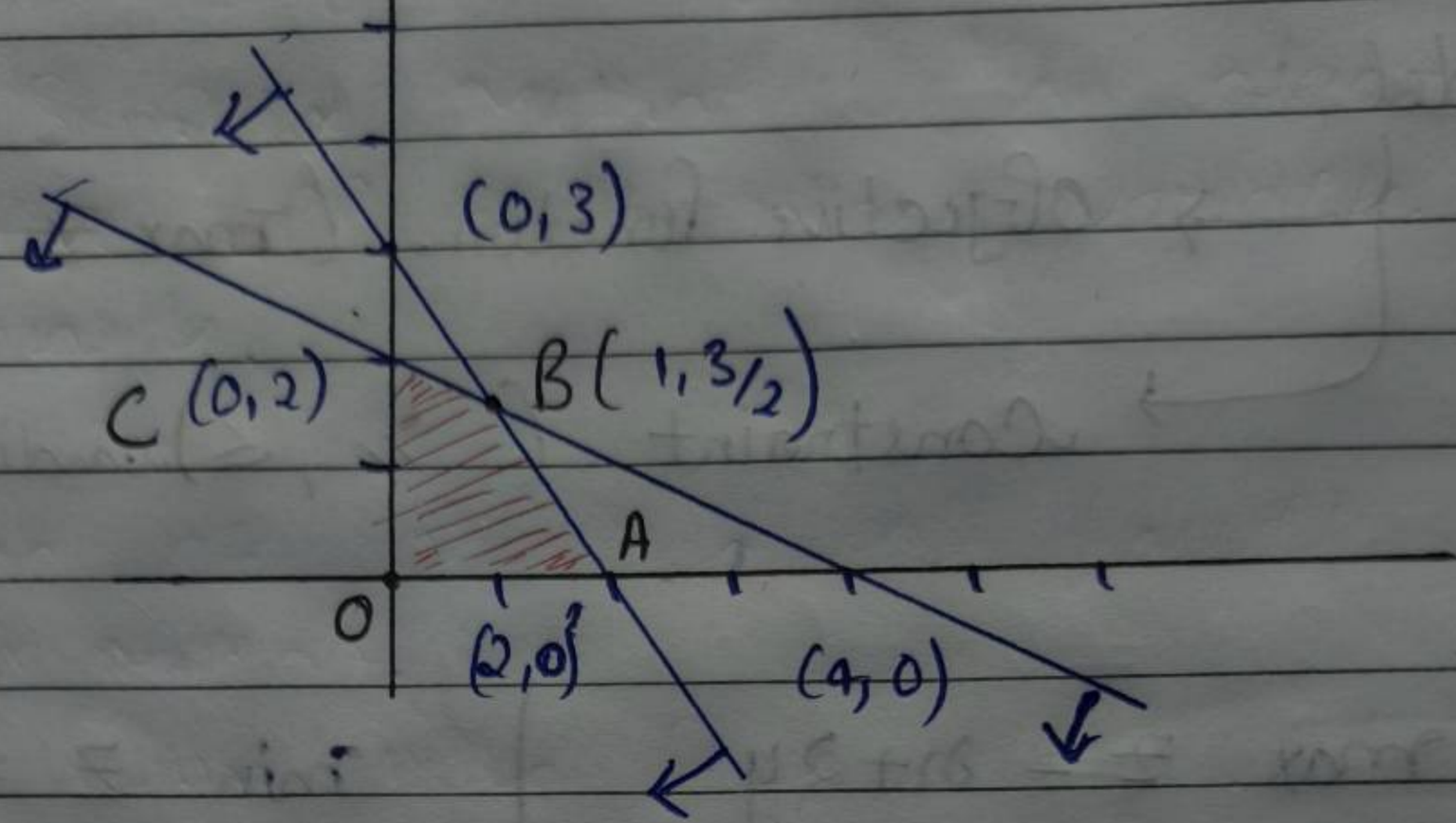
$$x_1 + 2x_2 \leq 4 \quad \text{—— (i)}$$
$$3x_1 + 2x_2 \leq 6 \quad \text{—— (ii)}$$
$$x_1, x_2 \geq 0$$

for (i)

x_1	4	0
x_2	0	2

for (ii)

x_1	2	0
x_2	0	3



$x_1 + 2x_2 = 4$

$3x_1 + 2x_2 = 6$

$x_1 = 4 - 2x_2$

$3(4-2x_2) + 2x_2 = 6$

$12 - 6x_2 + 2x_2 = 6$

$6 = 4x_2$

$\boxed{x_2 = \frac{3}{2}}$

$\rightarrow x_1 = 4 - \cancel{2} \times \frac{3}{\cancel{2}}$

$\boxed{x_1 = 1}$

A(2,0)

$\Rightarrow$ B(1, 3/2)

C(0, 2)

O(0, 0)

O	A	B	C
2	0	6	9	8

So,

It is optimal sol^n at $x_1 = 1$, $x_2 = 3/2$.

$\boxed{\max z = 9}$

Q min $Z = 2x_1 + 3x_2$

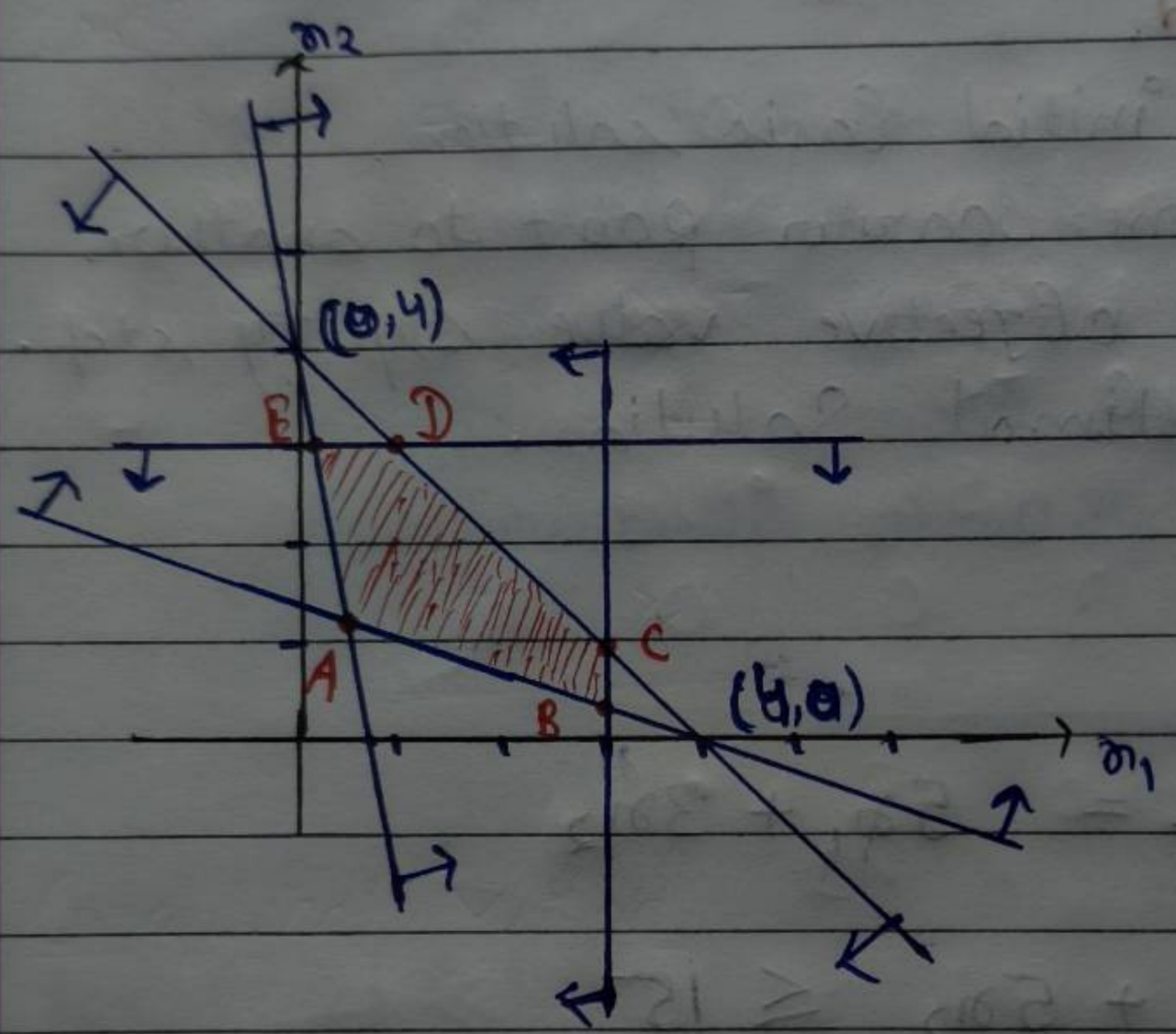
$x_1 + x_2 \leq 4$ — (i)

$6x_1 + 2x_2 \geq 8$ — (ii)

$x_1 + 5x_2 \geq 4$ — (iii)

$x_1 \leq 3, \quad x_2 \leq 3$

$x_1 \geq 0, \quad x_2 \geq 0$



for eq (i)

x_1	0	4
x_2	4	0

for eq (ii)

x_1	0	4/3
x_2	4	0

for eq (iii)

x_1	0	4
x_2	4/5	0

A (8/7, 4/7)
B (3, 1/5)
C (3, 1)
D (1, 3)
E (1/3, 3)

at	A	B	C	D	E
Z	4	6.6	9	11	9.66

So, the optimal solution for

$\boxed{\min Z = 4}$

at A.

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Simplex Method

The Simplex Method is an algorithm used to solve LPP with many variable and constraints.

Instead of checking all possible solution, Simplex Method.

→ Start form initial basic solution.
→ Moves from one corner point to another.
→ Improves the objective value step by step.
→ Stop at optimal Solution.

Ex:-

$$\max Z = 5x_1 + 3x_2$$

$$3x_1 + 5x_2 \leq 15$$
$$5x_1 + 2x_2 \leq 10$$
$$x_1, x_2 \geq 0$$

Adding of Slack & Surplus variable.

- Slack :-

→ It converts ≤ constraint into equation.

Ex:-

$$x + y \leq 5$$

Add Slack variable. S_1

→ $x + y + S_1 = 5$

where

$S_1 \geq 0$.

{ Slack variable represents unused resources. }

- **Surplus :-** Used when constraint is $\geq$

Ex :-

$$x + y \geq 5$$

Subtract :-

$$x + y - S_1 = 5$$

Surplus variable represents excess production.

Then,

$$\max Z = 5x_1 + 3x_2 + 0.x_3 + 0.x_4$$

$$3x_1 + 5x_2 + x_3 + 0.x_4 = 15$$
$$5x_1 + 2x_2 + 0.x_3 + x_4 = 10$$

$$x_i \geq 0$$

		C_j	5	3	0	0	
C_B	X_B	b	x_1	x_2	x_3	x_4	Min Ratio
0	x_3	15	3	5	1	0	$15/3 = 5$
0	x_4	10	5	2	0	1	$10/5 = 2$
	Z_j		0	0	0	0	
	C_j		5	3	0	0	
$Z_j - C_j$			-5	-3	0	0	

$b \rightarrow$ Solution

$C_j \rightarrow$ Cost of variable.

$X_B \rightarrow$ Basic Var

$C_B \rightarrow$ Cost of Basic variable

★ $\boxed{Z_j = C_B \times x_i}$

Take most -ve

↳ find Ratio (mini)

↳ Substitute basic variable with x_i of most -ve

As Above,

Min Ratio → 2

$x_4 \rightleftharpoons x_1$ & $5 \rightarrow 1$, $3 \rightarrow 0$.

(Divide by 5)

Old line $- 3 \times$ new line.

		C_j	5	3	0	0	
C_B	X_B	b	x_1	x_2	x_3	x_4	Mini Ratio
0	x_3	9	0	19/5	1	-3/5	45/19 = 2.... ← outgoing
5	x_1	2	1	2/5	0	1/5	10/2 = 5
	Z_j		5	2	0	1	
	C_j		5	3	0	0	
	$Z_j - C_j$		0	-1	0	1	

Old $- \frac{2}{5} \times$ New.

		C_j	5	3	0	0
C_B	X_B	b	x_1	x_2	x_3	x_4
3	x_2	45/19	0	1	5/19	-3/19
5	x_1	20/19	1	0	-2/19	5/19
	Z_j		5	3	5/19	16/19
	C_j		5	3	0	0
	$Z_j - C_j$		0	0	5/19	16/19

As,

All $Z_j - C_j \geq 0$ (Optimal solution condition)

$$x_1 = \frac{20}{19}, \quad x_2 = \frac{45}{19}$$

$$\max Z = 5x_1 + 3x_2$$

$$\Rightarrow 5 \times \frac{20}{19} + 3 \times \frac{45}{19}$$

$$\Rightarrow \frac{235}{19}$$ Ans.

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Q $\max Z = 3x_1 + 2x_2 + 5x_3$

$$x_1 + x_2 + x_3 \le 9$$
$$2x_1 + 3x_2 + 5x_3 \le 30$$
$$2x_1 - x_2 - x_3 \le 8$$
$$x_i \ge 0$$

$$\max Z = 3x_1 + 2x_2 + 5x_3 + 0.x_4 + 0.x_5 + 0.x_6$$

$$x_1 + x_2 + x_3 + x_4 + 0.x_5 + 0.x_6 = 9$$
$$2x_1 + 3x_2 + 5x_3 + 0.x_4 + x_5 + 0.x_6 = 30$$
$$2x_1 - x_2 - x_3 + 0.x_4 + 0.x_5 + x_6 = 8$$

$$x_i \ge 0$$

		C_j	3	2	5	0	0	0	
C_B	X_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
0	x_4	9	1	1	1	1	0	0	$9/1 = 9$
0	x_5	30	2	3	5	0	1	0	$30/5 = 6$ ← outgo
0	x_6	8	2	-1	-1	0	0	1	—
	Z_j		0	0	0	0	0	0	
	C_j		3	2	5	0	0	0	
	$Z_j - C_j$		-3	-2	-5	0	0	0	

			3	2	5	0	0	0	
C_B	X_B	b	x_1	x_2	x_3	x_4	x_5	x_6	min ratio
0	x_4	3	$3/5$	$2/5$	0	1	$-1/5$	0	$3/(3/5) = 5$ ← outgoing
5	x_3	6	$2/5$	$3/5$	1	0	$1/5$	0	$6/(2/5) = 15$
0	x_6	14	$12/5$	$-2/5$	0	0	$1/5$	1	$14/(12/5) = 35/6 = 5.8$

z_j	2	3	5	0	1	0
c_j	3	2	5	0	0	0
$z_j - c_j$	(−1)	1	0	0	1	0

		c_j	3	2	5	0	0	0
C_B	X_B	b	x_1	x_2	x_3	x_4	x_5	x_6
3	x_1	5	1	2/3	0	5/3	−1/3	0
5	x_3	4	0	1/3	1	−2/3	1/3	0
0	x_6	2	0	−2	0	−4	1	1
	z_j		3	11/3	5	5/3	2/3	0
	c_j		3	2	5	0	0	0
	$z_j - c_j$		0	5/6	0	5/3	2/3	0

$\boxed{\text{old} - c \cdot \text{new}}$

$\boxed{\text{old} + c \cdot \text{new}}$

So,

All $z_j - c_j \geq 0$

$x_1 = 5$, $x_3 = 4$, $x_2 = 0$

$$\max z = 3x_1 + 2x_2 + 5x_3$$
$$\Rightarrow 3 \times 5 + 2 \times 0 + 5 \times 4$$
$$\Rightarrow 15 + 0 + 20$$
$$\Rightarrow 35 \quad \text{Ans}$$

Big - M method :

→ Another method to deal with artificial variables.

→ Artificial variables are added with very large penalty M.

Objective function becomes :-

$$Z = 3x_1 + 2x_2 - Ma_1 - Ma_2$$

M ⇒ very large number
a_1, a_2 ⇒ artificial variables.
Z → always be maximize.

Q $\min Z = x_1 + x_2$.

s.t.,

$$2x_1 + x_2 \geq 4$$
$$x_1 + 7x_2 \geq 7$$
$$x_i \geq 0$$

min Z ⟶ max Z by multiply (-1) both side.

$$\max Z = -x_1 - x_2 + 0.x_3 + 0.x_4 - Mx_5 - Mx_6$$

$$2x_1 + x_2 + x_3 + 0.x_4 + x_5 + 0.x_6 = 4$$
$$x_1 + 7x_2 + \underbrace{0.x_3 + x_4}_{\text{Surplus variable}} + \underbrace{0.x_5 + x_6}_{\text{artificial variable}} = 7$$

$x_i \geq 0$

		C_j	-1	-1	0	0	-M	-M	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Minratio
-M	x_5	4	2	1	-1	0	1	0	$4/1 = 4$
-M	x_6	7	1	7	0	-1	0	1	$7/7 = 1$ ←
	Z_j		-3M	-8M	M	M	-M	-M	
	C_j		-1	-1	0	0	-M	-M	
	$Z_j - C_j$		-3M+1	-8M+1	M	M	0	0	

↑

		C_j	-1	-1	0	0	-m	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Minratio
-M	x_5	3	13/7	0	-1	1/7	1	$3/(13/7) = 21/13$ ← Min
-1	x_2	1	1/7	1	0	-1/7	0	$1/(1/7) = 7$
	Z_j		$-\frac{13}{7}m - \frac{1}{7}$	(-1)	M	$\frac{-M+1}{7}$	-M	
	C_j		-1	-1	0	0	-M	
	$Z_j - C_j$		$\underbrace{-\frac{13}{7}m + \frac{6}{7}}_{-ve}$	0	M	$-\frac{M+1}{7}$	0	

		C_j	-1	-1	0	0	
C_B	x_B	b	x_1	x_2	x_3	x_4	Minratio
-1	x_1	21/13	1	0	-7/13	1/13	
-1	x_2	10/13	0	1	1/13	-2/13	
	Z_j		-1	-1	6/13	1/13	
	C_j		-1	-1	0	0	
	$Z_j - C_j$		0	0	6/13	1/13	

↳ All +ve.

So,

$$x_1 = \frac{21}{13}, \quad x_2 = \frac{10}{13}$$

$$\min z = x_1 + x_2 = \frac{21}{13} + \frac{10}{13} = \frac{31}{13}.$$

~~Q~~ $Z = 5x_1 + 2x_2 + 8x_3$

~~S.t.~~

$$2x_1 - 4x_2 + x_3 \le 42$$
$$2x_1 + 3x_2 - x_3 \le 42$$
$$6x_1 - x_2 + 3x_3 \le 42$$
$$x_i \ge 0$$

~~max~~ $Z = 5x_1 + 2x_2 + 8x_3$

$$2x_1 - 4x_2 + x_3 + x_4 + 0.x_5 + 0.x_6$$
$$2x_1 - 3x_2 - x_3 + 0.x_4 + x_5 + 0.x_6$$
$$6x_1 - x_2 + 3x_3 + 0.x_4 + 0.x_5 + x_6$$

Q $Z = 6x_1 - 9x_2$

By Big M

S.t. :-

$$2x_1 - 3x_2 \le 6$$
$$x_1 + x_2 \le 20$$
$$x_i \ge 0$$

$$\max Z = 6x_1 - 9x_2 + 0.x_3 + 0.x_4 - Mx_5 - Mx_6$$

$$2x_1 - 3x_2 + x_3 + 0.x_4 + x_5 + 0.x_6 = 6$$
$$x_1 + x_2 + 0.x_3 + x_4 + 0.x_5 + x_6 = 20$$

		C_j	6	-9	0	0	-M	-M	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Min
-M	x_5	6	2	-3	1	0	1	0	6/2 = 3 ←
-M	x_6	20	1	1	0	1	0	1	20/1 = 20
	Z_j		-3M	2M	-M	-M	-M	-M	
	C_j		6	-9	0	0	-M	-M	
	$Z_j - C_j$		-3M-6 ↑	2M+9	-M	-M	0	0	

		C_j	6	-9	0	0	-M	
C_B	X_B	b	x_1	x_2	x_3	x_4	x_6	Mini
6	x_1	3	1	$-3/2$	$1/2$	0	0	$3/3/2 = -2$ ✗
-M	x_6	17	0	$5/2$	$-1/2$	1	1	$17/5/2 = 34/5$ ←
	Z_j		6	$-9-5M/2$	$3+M/2$	$-M$	$-M$	
	C_j		6	-9	0	0	-M	
			0	$-5M/2$	$3+M/2$	$-M$	0	

↑

C_B	x_B	b	x_1	x_2	x_3	x_4
6	x_1	$66/5$	1	0	$1/5$	$3/5$
-9	x_2	$34/5$	0	1	$-1/5$	$2/5$
	Z_j		6	-9	3	0
	C_j		6	-9	0	0
$Z_j - C_j$			0	0	3	0

$Z_j - C_j \geq 0$ (All +ve)

$x_1 = \frac{66}{5}$, $x_2 = \frac{34}{5}$

$$\text{Max } Z = 6x_1 - 9x_2$$

$$\Rightarrow 6 \times \frac{66}{5} - 9 \times \frac{34}{5}$$

$$\Rightarrow \frac{396}{5} - \frac{306}{5}$$

$$\Rightarrow \frac{90}{5} = 18$$

$\boxed{\text{Max } Z = 18}$ Ans

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Two-Phase Method :-

→ It is used when initial basic feasible solution is not obvious.

It solves LPP in two Phases :-

Phase 1 :-

→ Introduce Artificial variables.
→ Minimize the sum of Artificial variables.
→ find a feasible solution.

Phase 2 :-

→ Remove artificial variables.
→ Solve the original LPP using Simplex Method.

Ex :-

$$\min z = x_1 + x_2.$$

Sol:,

$$2x_1 + x_2 \geq 4$$
$$x_1 + 7x_2 \geq 7$$
$$x_i \geq 0$$

(i) convert in max z,

$$\max z = -x_1 - x_2.$$

$$2x_1 + x_2 - x_3 + 0.x_4 + x_5 + 0.x_6 = 4$$
$$x_1 + 7x_2 + 0.x_3 - x_4 + 0.x_5 + x_6 = 7$$

Phase I :-

$$\max Z = 0.x_1 + 0.x_2 + 0.x_3 + 0.x_4 - x_5 - x_6$$

		C_j	0	0	0	0	-1	-1	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Mini ratio
-1	x_5	4	2	1	-1	0	1	0	$4/1 = 4$
-1	x_6	7	1	7	0	-1	0	1	$7/7 = 1$
	Z_j		-3	-8	1	1	-1	-1	
	C_j		0	0	0	0	-1	-1	
	$Z_j - C_j$		-3	-8	1	1	0	0	

		C_j	0	0	0	0	-1	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Mini-
-1	x_5	3	13/7	0	-1	1/7	1	$3/(13/7) = 21/13$
0	x_2	1	1/7	1	0	-1/7	0	$1/(1/7) = 7$
	Z_j		-13/7	0	1	-1/7	-1	
	C_j		0	0	0	0	-1	
	$Z_j - C_j$		-13/7	0	1	-1/7	0	

		C_j	0	0	0	0	
C_B	X_B	b	x_1	x_2	x_3	x_4	Mini.
0	x_1	21/13	1	0	-7/13	1/13	
0	x_2	10/13	0	1	1/13	-2/13	
	Z_j		0	0	0	0	
	C_j		0	0	0	0	
	$Z_j - C_j$		0	0	0	0	

All zero.

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Phase 2 :-

$$\max Z = -x_1 - x_2 + 0x_3 + 0x_4$$

		C_j	-1	-1	0	0
C_B	x_B	b	x_1	x_2	x_3	x_4
-1	x_1	$21/13$	1	0	$-7/13$	$1/13$
-1	x_2	$10/13$	0	1	$1/13$	$-2/13$
	z_j		-1	-1	$6/13$	$1/13$
	c_j		-1	-1	0	0
	$z_j - c_j$		0	0	$6/13$	$1/13$

All +ve

So,

$$x_1 = \frac{21}{13}, \quad x_2 = \frac{10}{13}.$$

$$\min z = x_1 + x_2 = \frac{21}{13} + \frac{10}{13} = \frac{31}{13}$$

Ans

Q $Z = x_1 + 2x_2$

By Two Phase

Sol,

$$x_1 + 3x_2 \le 15$$
$$2x_1 - x_2 \le 12$$
$$x_i \ge 0$$

$$\text{max } Z = x_1 + 2x_2 + 0.x_3 + 0.x_4 + 0.x_5 + 0.x_6$$

$$x_5 + 0.x_6 + x_1 + 3x_2 + x_3 + 0.x_4 = 15$$
$$0.x_5 + x_6 + 2x_1 - x_2 + 0.x_3 + x_4 = 12$$

Phase (1) :-

$$\text{max } Z = 0x_1 + 0x_2 + 0x_3 + 0x_4 - x_5 - x_6$$

		C_j	0	0	0	0	-1	-1	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Minratio
-1	x_5	15	1	3	1	0	1	0	$15/1 = 15$
-1	x_6	12	2	-1	0	1	0	1	$12/2 = 6$ ←
	Z_j		-3	-2	-1	-1	-1	-1	
	C_j		0	0	0	0	-1	-1	
$Z_j - C_j$			-3 ↑	-2	-1	-1	0	0	

		C_j	0	0	0	0	-1	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Min.
-1	x_5	9	0	7/2	1	-1/2	1	18/7 ←
0	x_1	6	1	-1/2	0	1/2	0	—
	Z_j		0	-7/2	-1	1/2	-1	
	C_j		0	0	0	0	-1	
			0	-7/2 ↑	-1	1/2	0	

C_B	x_B	b	x_1	x_2	x_3	x_4
0	x_2	$18/7$	0	1	$2/7$	$-1/7$
0	x_1	$51/7$	1	0	$1/7$	$3/7$
	Z_j		0	0	0	0
	C_j					

All zero.

Phase ②

$$\max Z = x_1 + 2x_2 + 0.x_3 + 0.x_4.$$

		C_j	1	2	0	0	
C_B	x_B	b	x_1	x_2	x_3	x_4	Mini
2	x_2	$18/7$	0	1	$2/7$	$-1/7$	—
1	x_1	$51/7$	1	0	$1/7$	$3/7$	$\frac{51}{7} \times \frac{7}{1} = 102$
	Z_j		1	2	$5/7$	$10/7$	
	C_j		1	2	0	0	
			0	0	$-5/7$	$-10/7$	

↳ All zero

So,

$$\text{Max } Z = x_1 + 2x_2$$
$$\Rightarrow \frac{51}{7} + \frac{18}{7} \times 2$$
$$\Rightarrow \frac{87}{7}$$

Ans.

Duality of LPP :- ★★ 2.5 Marks Question

$$\max Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n.$$

S.t.

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \le b_1 \quad (y_1)$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \le b_2 \quad (y_2)$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \le b_n \quad (y_n)$$

$$\forall\, x_i \ge 0, \quad i = 1, 2, \cdots n.$$

4 → new variable.

Convertion from $\max Z \Longrightarrow \min \omega$

$$\min \omega = b_1y_1 + b_2y_2 + \cdots + b_ny_n.$$

S.t.

$$a_{11}y_1 + a_{12}y_2 + \cdots + a_{1n}y_n \ge c_1$$
$$a_{21}y_1 + a_{22}y_2 + \cdots + a_{2n}y_n \ge c_2$$
$$\vdots$$
$$a_{n1}y_1 + a_{n2}y_2 + \cdots + a_{nn}y_n \ge c_n$$

$$\forall\, y_i \ge 0, \quad i = 1, 2, \cdots n.$$

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Q $\max \; Z = 2x_1 + 3x_2$

St,

$$x_1 + x_2 \leq 4 \qquad (y_1)$$
$$x_1 - x_2 \leq 1 \qquad (y_2)$$
$$x_i \geq 0$$

$\min \; \omega = 4y_1 + y_2.$

S.t.,

$$y_1 + y_2 \geq 2$$
$$y_1 - y_2 \geq 3 \qquad y_i \geq 0, \; i = 1, 2$$

Ans

Q $\max Z = 5x_1 + 2x_2 + 8x_3$

St;

$$2x_1 - 4x_2 + x_3 \leq 42 \qquad (y_1)$$
$$2x_1 + 3x_2 - x_3 \leq 42 \qquad (y_2)$$
$$6x_1 - x_2 + 3x_3 \leq 42 \qquad (y_3)$$
$$x_1, x_2, x_3 \geq 0$$

$\min \; \omega = 42y_1 + 42y_2 + 42y_3$

S.t.,

$$2y_1 - 4y_2 + y_3 \geq 5$$
$$2y_1 + 3y_2 - y_3 \geq 2$$
$$6y_1 - y_2 + 3y_3 \geq 8$$

$$y_i \geq 0, \quad i = 1, 2, 3.$$